A Basis for the Dyadic Sections of Euler's Totient

A structure theorem, a sharp Farey denominator bound, and the exact boundary of a kernel-checked attack on Erdős #249

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Abstract

Erdős #249 asks whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational. It is open, nothing here closes it, and this note reports what a kernel-checked attack does yield, each claim bounded by the exact proposition a proof checker accepted — the negative results included.

For $j \geq 0$ and $0 \leq r < 2^j$ let $\varphi_{j,r}\colon n \mapsto \varphi(2^j n + r)$. Main theorem: $\varphi_{0,0}$, $\varphi_{1,0}$ and the sections $\varphi_{j,r}$ with r odd form a $\mathbb{Q}$-basis of the span of *all* the $\varphi_{j,r}$, so that span is infinite-dimensional, its level-e part has dimension exactly $2^e + 1$, and the only $\mathbb{Q}$-linear relations among dyadic sections of φ are those generated by two elementary identities.

Second, $S \neq a/q$ for every $a \in \mathbb{Z}$ and every $q \leq 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053$, a Farey exclusion whose method is standard: we name it first, claim no novelty, and add only that the constant is *exactly optimal* for its window. Third, the Lambert weight $A = \varphi * \mu$ of S is unbounded and no fixed positive integer clears its coordinates $A(n)/n$; that is what separates #249 from three settled neighbours. Every equivalence below is labelled one: an equivalence is not progress.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

1 Results

Every item is a proposition the pinned Lean kernel accepted, or (R9) the problem itself; the tag gives its status in the claim registry's taxonomy, and each *Checked* line links the declarations at commit `571ec44f2aad`. Nothing in the list is an irrationality criterion for S, and no combination of the items becomes one. Throughout $\varphi_{j,r}(n) = \varphi(2^j n + r)$, for $j \geq 0$ and $0 \leq r < 2^j$, is the (j,r) *dyadic section* of Euler's totient, a vector in $\mathbb{Q}^{\mathbb{N}}$. Two elementary identities relate them: the *reductions* $\varphi_{j+1,0} = 2^j \varphi_{1,0}$ (checked) and $\varphi_{j,2^{t+1}s} = 2^t \varphi_{j-t-1,s}$ for s odd with $2^{t+1}s < 2^j$ (checked).

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R1. [PROVED HERE] **Basis theorem.** The family $\mathcal{B} = \{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1,\ r \text{ odd},\ 0 < r < 2^j\}$ is linearly independent over $\mathbb{Q}$ and spans the same $\mathbb{Q}$-subspace of $\mathbb{Q}^{\mathbb{N}}$ as the set of *all* dyadic sections $\{\varphi_{j,r} : j \geq 0,\ 0 \leq r < 2^j\}$; so $\mathcal{B}$ is a basis of that span, which is therefore infinite-dimensional. Since the reductions make every remaining section a rational multiple of a member of $\mathcal{B}$, the $\mathbb{Q}$-linear relations among the dyadic sections of φ are exactly those the reductions generate — a one-line consequence of the checked statements, not a further one. *Checked:* independence, span equality, basis.

R2. [PROVED HERE] **Exact finite-level rank.** For every $e \geq 0$ the $2^e + 1$ sections $\varphi_{0,0}$, $\varphi_{1,0}$ and $\varphi_{j,r}$ with $1 \leq j \leq e$, r odd, $0 < r < 2^j$ are linearly independent over $\mathbb{Q}$, so their span has dimension exactly $2^e + 1 = 2 + \sum_{j=1}^{e} 2^{j-1}$: exact, not an estimate, because the even residues are already dependent by the reductions. R1 follows by finite character of linear independence. *Checked:* independence, dimension.

R3. [UNCONDITIONAL PROGRESS] **Denominator exclusion, sharp for its window.** If $S = a/q$ with $a \in \mathbb{Z}$ and $q \geq 1$ then $q > 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$. The constant is optimal for the window producing it: the next integer up is the exact first denominator at which that window's certificate fails. The method is Farey (Section 4). *Checked:* exclusion, reduced-denominator form, first failure.

R4. [UNCONDITIONAL PROGRESS] **Carry anti-compression.** If S were rational then some tempered integral carry orbit for the coefficients φ, with positive multiplier, would have for *every* e carry sections through level e spanning a space of dimension at least $2^e - 1$. *Closes:* every rationality proof that would exhibit a carry of bounded section rank. *Does not close:* anything else — it is not an irrationality criterion, because no finite-rank upper bound is proved for the carries that rationality supplies. *Checked:* the barrier, rank transport from R2.

R5. [PROVED HERE] **The #249 weight.** $S = \sum_{d \geq 1} A(d)/(2^d - 1)$ with $A = \varphi * \mu$; here $A \geq 0$, $A(p) = p - 2$, $A(p^k) = \varphi(p^k) - \varphi(p^{k-1})$, and A is unbounded. *Checked:* identity, sign, primes, prime powers, unboundedness.

R6. [FORMALISED HERE] **Eventually periodic weights are settled.** For every integer base $b \geq 2$, if $\gamma\colon \mathbb{N} \to \mathbb{Q}$ is eventually periodic, nonnegative and positive at some positive index of its periodic part, then $\sum_a \gamma(a)/(b^a - 1)$ is irrational (Luca and Tachiya prove a broader periodic theorem by other methods [3]; no priority is claimed). By R5 the weight A is unbounded, hence not eventually periodic, so #249 lies just outside this class. *Checked:* the periodic theorem.

R7. [PROVED HERE] **No fixed index clears the normalised weight.** For every integer $D \geq 1$ some $n \geq 1$ has $D \cdot A(n)/n \notin \mathbb{Z}$, and any D clearing every coordinate up to a horizon $N \geq 4$ is divisible by an explicit two-tier primorial (4 at $p = 2$, p^2 when $p^2 \leq N$, else p). *Closes:* every argument clearing the coordinates $A(n)/n$ by a single positive integer valid at all n. *Does not close:* arguments whose index grows with n, or which never clear the coordinates. *Checked:* no fixed index, the primorial divisibility.

R8. [CONDITIONAL REDUCTION] **Two exact reformulations, offered as nothing more.** Irrationality of S is equivalent to a pointwise finite-certificate condition, and follows from a supply of gap certificates — one window (N, K) per denominator q whose totient carry residue avoids a band of width $q(N+K+2)$ out of 2^K. Neither hypothesis is proved for one unbounded family. Both are restatements of #249 in other coordinates, and are the part of this work most likely to be mistaken for a result. *Checked:* the equivalence, the supply implication.

R9. [OPEN] **Erdős #249.** Whether S is irrational. Not proved here.

2 Where #249 sits

If $f = g * \mathbf{1}$, that is $f(n) = \sum_{d|n} g(d)$, then interchanging the two sums gives

$$\sum_{n\geq 1} \frac{f(n)}{2^n} = \sum_{d\geq 1} \frac{g(d)}{2^d - 1}. \tag{1}$$

Four classical coefficient sequences have this shape. The table records the constant, the weight $g = f * \mu$, and its status. Decimals were recomputed to 50 places for this note and are truncated, not rounded.

f	$\sum f(n)/2^n$	weight $g = f * \mu$	status
τ	$1.60669515\ldots = E$	$g \equiv 1$	Irrational (Erdős 1948); formalised here in the Lambert form $\sum_{d\geq 1}(2^d - 1)^{-1}$ (checked), equal to $\sum \tau(n)/2^n$ by (1)
ω	$0.51694281\ldots$	$g = \mathbf{1}_{\text{primes}}$	Irrational (Tao–Teräväinen 2025) [1]
Ω	$0.58950329\ldots$	$g = \mathbf{1}_{\text{prime powers}}$	Irrational (*ibid.*, remark)
φ	$1.36763080\ldots = S$	$g = A = \varphi * \mu$	**Open** (Erdős #249)

The first three weights are bounded, taking only the values 0 and 1 (for τ the weight is the constant 1, so only the value 1). The fourth is unbounded, by R5. That difference is the one that matters for the route in R6, and we do not claim it is the only difference: A also vanishes on $n \equiv 2 \pmod 4$, while $\mathbf{1}_{\text{primes}}$ is bounded but no more eventually periodic than A is, so the settled ω and Ω cases are not instances of R6 either. They were settled by a different method, discussed in Section 6.

The difficulty of #249 does not live in the Lambert transform. The neighbouring rungs are rational and exactly computable — $\sum_{d\geq 1} \mu(d)/(2^d - 1) = \frac{1}{2}$ (checked) and $\sum_{d\geq 1} \varphi(d)/(2^d - 1) = 2$ (checked) — and the signed squared-Mersenne lens $S = \frac{1}{2} + \sum_{d\geq 1} \mu(d)/(2^d - 1)^2$ (checked) recomputes S to all 50 places. The difficulty lives on the power-series side, and R1 is a statement about exactly that side.

3 The basis theorem

The producer for R2 is a separated evaluation minor. For each level, the Chinese remainder theorem together with Dirichlet's theorem on primes in arithmetic progressions supplies a set of evaluation points at which the section matrix has a nonzero minor with the required two-adic separation: one affine channel is made prime, and every other channel receives a fresh prime divisor congruent to 1 modulo a large power of two, so the columns carry distinct exact powers of two and the determinant cannot vanish (checked). Even residues never enter, because the second reduction has already carried them to odd-core channels (checked).

R1 then needs only two further steps, and both are short. Linear independence is a property of finite subfamilies, so the level-e producer, applied with e above the largest level occurring in a given finite subset, already gives independence of the whole infinite family $\mathcal{B}$. And the two reductions carry every remaining section onto a rational multiple of a member of $\mathcal{B}$, so the spans agree. That the assembly is short is the point: the level-e theorem was always the whole content, and stating its consequence as infinite-dimensionality understated it. The sharp form is a basis.

Remark. R1 and R2 are statements about φ, not about S; they are true, and their proofs are complete, whether or not S is irrational. They are here because they are the exact form of "no finite amount of the totient's dyadic structure determines the rest" — the property that R6 consumes for an eventually periodic weight and that A lacks.

The rationality side. Binary long division turns a hypothetical rational value into an integer carry recurrence: for coefficients $c(n) \leq n$, and in particular for $c = \varphi$, the series $\sum c(n)/2^n$ is rational exactly when a tempered integer carry orbit exists (checked). Transporting R2 through that recurrence gives R4. The scope line there is worth repeating: this is an anti-compression barrier, and the complementary finite-rank upper bound, which is what would make it a proof, is not available.

4 The denominator bound

R3 is a Farey exclusion, and we say so before saying anything else about it. The argument is the classical mediant one. The Lean lemma commits the first 240 binary digits of the shifted series as a single residue over the totient carry window $(N, K) = (1, 240)$; a rational a/q can equal S only if it falls into a resulting bad interval of width $243/2^{240}$; that interval is bracketed by two explicit unimodular fractions $a_1/b < c_1/d$ with $bc_1 - a_1 d = 1$; and the mediant lemma (checked) forces any rational strictly between unimodular neighbours to have denominator at least $b + d$. The excluded range is therefore $q \leq b + d - 1$, which is the printed constant (checked), and the transport to the series is a tail estimate (checked). *The method is standard, and we claim no novelty for it.*

Two things should be read off correctly. First, the scale. A window of K computed binary digits yields a mediant bound of order $2^{K/2}$: the same pipeline gives 2.49×10^{17} at $K = 120$ (checked) and 7.96×10^{34} at $K = 240$. The constant is a function of how many digits were committed, not a measure of how much the problem has moved. Second, the sharpness. Within its window the bound cannot be improved at all: $b + d$ itself fails the certificate, so $b + d - 1$ is exactly the last denominator excluded (R3). A reader who regards an explicit Farey exclusion as routine is not in disagreement with this note. R1, not R3, is where we would ask a sceptical reader to look.

The bound transfers to the coprimality form of the constant with the constant halved. Writing the visible-coprime-pair series as $\sum_{\gcd(m,n)=1,\ m,n\geq 1} 2^{-(m+n)}$, one has $S = \frac{1}{2} + \sum_{\gcd(m,n)=1} 2^{-(m+n)}$ (checked), so that series has no representation a/d with $d \leq 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526$ (checked), that number being $(q_0 - 1)/2$ for the constant q_0 of R3. This series is the probability that two independent geometric fair-coin waiting times are coprime; the checked statements are the series identity and the series exclusion, and the probabilistic reading is a gloss on them rather than a formalised measure-theoretic statement.

5 Barriers, and two we do not state as theorems

R7 is the one method barrier this note states, and its scope lines are given with it in Section 1. It is worth a sentence of mechanism, because the mechanism is the whole of it: take a prime $p > \max(D, 2)$; then $A(p) = p - 2$ by R5, and $p \nmid D(p-2)$, so $D \cdot A(p)/p \notin \mathbb{Z}$. Being one line is a virtue here. The statement is about the weight the abstract already names, it needs no vocabulary this note has not defined, and it survives a hostile reading.

Two further no-go results in the formal corpus appeared as propositions in an earlier version of this note, and are demoted rather than dropped, because the reason is more useful to a reader than the propositions were.

A fixed-precision transport result (checked) says that at every fixed positive precision, each finite compatible valuation-unit word admits a prefix-locked centred completion. Its statement quantifies over abstract pairs of a two-adic valuation and an odd unit; it mentions neither φ nor S, its content is that a congruence class meets any interval of the corresponding length, and it is a restatement of a lemma printed immediately above it. The class of arguments it closes — deriving a contradiction from a local signature read at a precision fixed in advance, along a finite word — is not one anybody has proposed for #249.

A factor-ideal result (checked, with a sparse-anchor companion at checked) exhibits, for each $t \geq 3$ with $H = \operatorname{lcm}(1, \ldots, t)$, a nonzero integer carry whose forcing letters lie in the ideal generated by $\varphi(H)$, which reproduces the true totient differences at $t - 2$ prescribed indices, and which every finite integer shift polynomial carries to a pair with the same coboundary form, the same ideal memberships and an ℓ^1-weight bound. It does close that hypothesis class. Two limits. The witness is a spike — its state is $-\varphi(H)$ at $t-2$ points and zero elsewhere, so it stays uniformly bounded while the strip bounds it respects grow, and the construction is cheap for that reason. And the shifted pairs are not asserted to be nonzero, so an argument that additionally demanded non-vanishing after the shift is not closed. An earlier draft of this note also attributed to it a whole-ray anchor condition, a diagonal-bound condition and a strict-survivor condition that the cited theorem does not contain, and described it as "surviving" the shift algebra when the checked conclusion carries no non-vanishing clause. Correcting that is the reason it is here at all.

6 What remains open

The problem is open and this note does not narrow it. Precisely:

1. Irrationality of S (R9). No proof is claimed.
2. The certificate routes of R8 convert irrationality into the existence of an unbounded supply of finite integer certificates. Finite instances are checked — R3 is one, at $K = 240$ — and the unbounded supply is not. *An equivalence is not progress.*
3. The natural analytic target — a cancellation estimate for the first-harmonic block sum attached to the doubling orbit of $(2^h - 1)S$ — is unproved, and is strictly stronger than irrationality rather than equivalent to it.

A route we would now take seriously, and do not have: adapt the Gowers-uniformity and quantitative-correlation method of [1] from the bounded weight $\mathbf{1}_{\text{primes}}$ to an unbounded weight. Their argument uses the prime-dilation identity $\omega(pn) = \omega(n) + 1 - \mathbf{1}_{p|n}$, whose main increment is constant and whose error is sparse. For φ the corresponding correction is valuation-dependent, and the formal corpus makes the difficulty precise rather than impressionistic: on any prescribed finite horizon the correction *can* be suppressed, by a bounded simultaneous square-CRT base (checked), but suppression alone forces nothing, since a clean block can vanish (checked) and can equally be nonzero (checked). So the transfer is not blocked by an obvious obstruction and is not routine either. It is the first place a reader who wants to close #249 should look, and it did not exist when this development began.

Statements and declarations

This manuscript is authored exposition, not proof authority. The linked Lean snapshot is authoritative only for its exact propositions, and kernel checking establishes that a proposition was proved, not that it is interesting, novel, or sufficient. The consequence drawn in the last sentence of R1, the deduction in R6 that an unbounded weight is not eventually periodic, and the mechanism sentence in Section 5 are one-line arguments in the prose, marked as such, from statements that are checked. Numerical decimals quoted in Section 2 were recomputed independently for this note. Erdős Problem #249 remains open.

References

[1] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Proves irrationality of $\sum_{n\geq 1} \omega(n)/2^n$, and remarks that the method also gives $\sum_{n\geq 1} \Omega(n)/2^n$ and every integer base $b \geq 2$.

[2] K. Pratt, *The irrationality of a prime factor series under a prime tuples conjecture*, arXiv:2409.15185. The conditional predecessor of the above.

[3] F. Luca and Y. Tachiya, *Irrationality of Lambert series associated with a periodic sequence*.

[4] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, 1980, p. 61.

[5] T. Bloom, *Erdős Problems*, `https://www.erdosproblems.com/249`.